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Recent achievements in electroanalysis of pharmaceuticals at carbon paste electrodes and related configurations

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ABSTRACT

In this review (with 167 references), recent achievements in the electroanalysis of pharmaceuticals with carbon paste electrodes (CPEs) and related configurations are presented. The text begins with a brief historical excursion, followed by a concise survey of the specific features of carbon paste-based electrodes. The CPEs reviewed are represented by mixtures from traditional components, i.e., powdered graphites and mineral oils, covering both unmodified and modified variants (except biosensors), as well as by alternative mixtures from new forms of carbon and/or other liquid binders. A pivotal part of this review presents a critical commentary with a series of tables gathering the important data on the modification and analytical performance of CPEs from original articles published in the last five years. All typical categories of pharmaceuticals, such as analgesics, antibiotics, anticancer and hormonal drugs, etc., are covered, together with excipients and impurities present in commercial formulations and real samples. Moreover, the possibilities and limitations of carbon pastes in electrochemical mechanistic studies of selected pharmaceuticals are also discussed. Finally, recent advances in the field and the most evident trends are highlighted, and some future prospects are given.

1. Introductory notes

It is almost three-quarters of a century since Adams shortly informed about the discovery of a previously unknown type of carbon electrode [1], the material of which had not been compact, but consisted of a mixture of graphite powder with a liquid organic binder. Due to its soft consistency, the author fittingly referred to it as "carbon paste" and the corresponding assembly in a suitable body as "carbon paste electrode" (CPE [1–3]). The new form of carbonaceous material had shown great promise and soon became popular among electrochemists and electroanalysts not only in the United States (see references in [2,3] and [4]), but also in Europe [5–7], as well as in the rest of the world ([8–10] and references therein). In the early era, the existing CPEs had been mainly made from spectroscopic graphite powders and paraffin oils [2,3] which are now called traditional or classic configurations, respectively. However, as time passed, the spectrum of carbon pastes gradually increased with mixtures prepared from other binders, such as silicone oils and greases (siloxane polymers [11]).

A fundamental step towards the truly global expansion of carbon paste electrodes was the introduction of chemically modified variants (abbreviated as CMCPEs) in the early 1980s ([8,9,12] and references therein). The basic two-component carbon paste contained another, usually minor component that desirably altered the properties of the resultant configurations. Shortly after, similarly designed biological modifications with enzymes in the role of effective (bio)modifiers emerged, classified as carbon paste biosensors (CP-biosensors [10]).

Then, the first mixtures of alternative carbon materials began to appear [13], and with the commencement of a new millennium, so-called new kinds of carbon took over this role; namely, nanotubes and graphene, both in a fine powder form ([14] and especially review [15] with citations therein). In parallel, other binders were tested in mixtures with traditional graphites, as well as with the aforementioned new carbons (e.g., lipophilic organic esters [16] and then some ionic liquids [17]). It is therefore logical that over the decades, a myriad of various carbon pastes — either modified or unmodified, or even altered otherwise — including the mentioned new mixtures (a.k.a. pseudo-carbon pastes or carbon paste-like composites), have been tested and the successful configurations further purposely employed.

The long period of measurements with CPEs and CMCPEs has been continuously mapped by review articles: from surveys of the very first findings from the laboratories of the inventor of carbon paste and his collaborators [2,3], through periodical overviews with a broader scope

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[8–10,15], up to the latest compilations [18–20] that already are more specialised. Finally, the entire era of CPEs, CMCPEs and CP-biosensors was compiled in a special monograph with more than three thousand citations [21], covering all activities with carbon paste-based electrodes and sensors within the period 1958–2012.

Regarding the properties and behaviour of carbon pastes, one should at least briefly review the specifics that are repeatedly emphasised in original papers [4–7,11–13,16] and compilations [2,3,8–10,15,18–21]. At this point, it applies to the following:

- *Easy and quick preparation* not only of the carbon pastes themselves but also of most electrode bodies for these mixtures (Fig. 1), thanks to which CPEs have probably become the most used type of electrode prepared manually in laboratories.
- *Comfortable and yet highly effective renewal of the electrode surface*. In this respect, CPEs far surpass almost all solid materials, including all carbon-based substrates, and only mercury-based electrodes may offer the same capability.
- *Variable modifications of carbon pastes.* Apparently, this most valuable feature of CPEs is due to its multi-componential nature and the completely different character of the two main components, as well as the respective methods of preparation, which can utilise a wide range of various procedures, including some unique approaches [8,15,21].

- *Extraction into the carbon paste bulk* enabled by the presence of a liquid binder as a process that can be combined with other principles of accumulation. Extraction into CPEs is a unique property, achievable at other electrodes only through the use of additional membranes.
- *Heterogeneous character* and electrochemical properties of carbon pastes are, with some limitations, very close to those of carbon inks for screen-printed electrodes (SPEs [15,21]). Because of this, manually prepared CPEs may serve as a testing platform for electrochemical assays prior to transfer to SPEs, which can be produced with the same properties in large quantities.

However, there are also some rather problematic features of carbon pastes:

- *Specific kinetics* at the CPE surface as a result of the presence of the liquid binder, which results in moderated electrode reactions of numerous organic redox pairs. This drawback can be reduced by electrolytic treatments. It complicates electrochemical studies of reaction mechanisms, and CPEs are not usually recommended for this purpose.

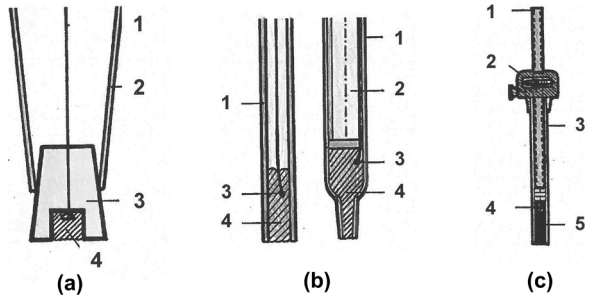


Fig. 1. Three common types of carbon paste holders and their construction. Schemes: a) "Filled plug" design according to Adams [3] constructed from Cu-wire with Pt-contact (1), glass tube (lower part conically narrowed) (2), Teflon® plug (3), and cavity filled with carbon paste (4). b) Glass and plastic tubes consisting of a tube or syringe (1), piston from glass rod (2), metal wire (3), and carbon paste filling (4). c) Piston-driven design [6] consisted of piston (1), turning head with screw (2), electrode body (3), Pt-contact (4), and carbon paste filling (5).

- Vulnerability of carbon pastes in *contact with organic solvents* with profound tendencies toward their total disintegration. Although some carbon paste mixtures can be stabilised by means of special modifiers, CPEs are not generally suitable for use in such media.
- *Worse reproducibility* of measurements due to the complex microstructure of the fragmented surface and, also, due to the manual renewal of the surface layer.
- Carbon paste is not compatible with spectroelectrochemistry because of its heterogeneous character and complex microstructure. In fact, in the entire era of the use of CPEs, there is only one report [22], where a compound of interest is dispersed in solid carbon paste and, in this form, is studied by CV and UV/Vis spectroelectrochemistry.
- Last but not least, there is a certain *unpredictability in the behaviour* of carbon pastes given by the type, quality, and mutual ratio of the main components, as well as the usual ageing effect of the carbon paste itself. All of this then requires experimental experience when working with CPEs, especially if newly proposed mixtures are to be used, which may discourage common users.

Before presenting the recent activities in pharmaceutical analysis with CPEs and CMCPEs in the period of 2020–2025, it is necessary to survey some basic facts regarding the electrochemistry of pharmaceuticals and related substances. It can be stated that this field, together with environmental analysis, represents the most effective application of electrochemical principles in practical measurements.

It can be traced in the literature that the electroanalysis of pharmaceuticals, as a separate subfield, had emerged in the second half of the 20th century [23,24] (in the case of classical polarography, it was even much earlier; see [25] and the references therein). Afterwards, pharmaceutical and clinical electroanalysis have been extensively mapped in numerous review articles (e.g., [26–28]) or book chapters [24,29], citing also contributions where CPEs have been used as working electrodes. Additionally, one review has also been devoted exclusively to electrodes made of chemically modified carbon pastes [27]. Later, actual advances have also been of interest [30–32], featuring some recent studies with CPEs and CMCPEs.

Regarding special monographs on pharmaceutical and clinical electroanalysis, one can point out a useful Wang's compilation from the late 1980s [33] or newer, and more widely focused books by renowned experts in the field [34,35]. In Ozkan's monograph [34], at the very beginning, a definition of what is understood by the term "pharmaceutical(s)" can be found, explaining why numerous contributions on pharmaceutical (electro)analysis deal with substances that are often classified separately within biologically important compounds.

The compilations mentioned above [24,26,34] have also summarised some typical features that, more or less, also apply to carbon paste-based electrodes:

- An indisputable benefit of electrochemical measurements is a very wide concentration range for numerous electroactive substances that can be reduced or oxidised. Modern electroanalytical instrumentation, combined with an optimally selected indicator/working electrode, enables the determination of concentrations ranging from the sub-nanomolar level to the millimolar level. On the one hand, this allows the detection of trace amounts of substances with high physiological activity, which can be administered even in extremely small amounts. However, analysis of commercial pharmaceutical preparations does not require achieving such low detection limits because the target analytes are present at much higher concentrations.
- In addition to directly oxidisable and reducible pharmaceuticals, there are also substances with a strong tendency to adsorb onto the electrode surface, either as such or in the form of complexes with suitable modifiers. Moreover, the extraction of neutral molecules into the electrode bulk or the formation of ion-pairs or associates is feasible, and these processes can also be employed in parallel. Thus,

these special approaches further expand the spectrum of substances to be determined.

- A great advantage of electroanalysis is the ability to distinguish between close derivatives by means of speciation or, at least, differentiation of two related forms.
- Simultaneous determination of some structurally different substances can also be carried out, if their redox characteristics or non-electrolytic interactions allow one to distinguish the corresponding analytical signals.
- Another specific feature of pharmaceutical electroanalysis is a certain predictability of eventual interferents, which usually applies to commercial formulations with almost known composition, here represented by the sample matrix.
- Electroanalytical experiments are well compatible with pre-concentration principles, which can be advantageously utilised in electrochemical stripping techniques. In this way, the substances monitored can be effectively accumulated on the surface of the working electrode or even in its bulk.
- Electroanalysis allows one to use different media for measurements (aqueous solutions, water-miscible solvents, or even mixtures of both), where the respective methods with suitably chosen electrodes resemble the corresponding procedures in RP-HPLC.
- A certain limiting factor may be the fact that the success of using electroanalytical techniques — and not only in pharmaceutical analysis — largely depends on an understanding of what is happening at the electrode/solution interface and what is measured and compared with standards and subsequently applied in quantitative analysis. Successful electroanalysis thus requires some experience and expertise, including a good understanding of electrochemical principles, as well as reliable performance of the electrodes and detectors used. This applies to a wider extent to chemically modified variants, which are continuously appearing in innovated or even completely new configurations.

Our original idea for this review was to prepare a continuation of a subchapter in the monograph on CPEs [21] devoted to pharmaceutical analysis and overviewing >200 different commercial drugs from the 1970s up to 2012. However, this intention had to be abandoned due to the finding that an excessive number of papers have appeared since 2013. Additionally, we decided to exclude the category of CP-biosensors, which is often included in numerous recent reviews concerning carbonaceous electrodes and pharmaceuticals (see e.g. [36–40]). Thus, our literature search has been narrowed down to the last five years, focusing on the configurations of CPEs and CMCPEs, as well as new types of carbon paste-like composites. All this was done in an effort to illustrate the state-of-the-art for the most recent era of pharmaceutical analysis with carbon paste-based electrodes in a critical rather than descriptive way.

2. Recent advances in electroanalysis of active pharmaceutical ingredients with carbon paste electrodes

Various active pharmaceutical ingredients were assayed at carbon paste electrodes using mostly pulse voltammetric techniques, such as differential pulse voltammetry (DPV) or square-wave voltammetry (SWV). In some cases, potentiometric sensors were also developed for that purpose. Generally, different modifications of the carbon paste were employed to propose the corresponding working electrodes with enhanced sensitivity and selectivity. Such sensors are typically designed to assay only one active pharmaceutical ingredient (API) of interest; however, there are numerous examples of modified CPEs that allow the simultaneous detection and quantification of multiple APIs in pharmaceutical formulations and also in biological fluids, such as blood, plasma, serum, or urine [41,42]. The analytical performance of modified CPEs in API assays can be characterised by detection limits even in nanomoles per litre [43–45]. However, in the majority of published methods, pharmaceuticals are reliably detected in the low micromolar range with a linear range spanning two orders of magnitude in concentration. The current applications reported in the last five years (Fig. 2) are summarised in Table 1, divided into the following sections according to the classes of APIs.

2.1. Analgesics

Most efforts were devoted to the development of modified CPEs for the electroanalysis of acetaminophen (paracetamol), a non-opioid analgesic and antipyretic agent. The modifiers were predominantly admixed directly into the carbon paste during its preparation, and loadings of up to 10 % (w/w) appear to be optimal; higher contents typically resulted in a decrease in the measured currents. Different types of clays, such as montmorillonite [46] or stevensite [47], carbon nanomaterials [41,48,49], or metal oxides [50], have been investigated for the bulk modification of carbon paste. Surprisingly, raw and thermally treated cork were also used in high concentrations (up to 50 %) as a sustainable additive for the preparation of a carbon paste [51]. For surface modification, the yellow PX4R dye was electropolymerised using cyclic voltammetry in 0.1 mol L^{-1} NaOH after 10 cycles between –0.6 V and 1.6 V [52].

Because the electrochemical oxidation of acetaminophen is pH-dependent, experimental conditions need to be optimised for sensitive analysis. Typically, acetaminophen is monitored at physiological pH [46,47]; however, the use of highly acidic [51] or basic [41] media has also been reported. This is advantageous for adjusting the peak potential of acetaminophen oxidation, for example, when simultaneous analysis with other pharmaceuticals is required. In such cases, acetaminophen can be quantified in mixtures with other non-steroidal anti-inflammatory drugs, e.g., together with diclofenac, naproxen, and aspirin, using MWCNTs-CPE and the respective chemometric evaluation employing partial least square regression and artificial neural network [41].

Generally, the developed carbon paste sensors were used for the rapid and low-cost assays of acetaminophen in tablets with a high drug content. In this case, the question arises as to what extent the addition of various modifiers is beneficial; the same analysis can be performed reliably in these samples using a bare CPE only. As already pointed out [53], investigations and the use of various modifiers often bring only minor enhancements to the analytical performance of sensors with no significant improvements in practical applicability. Some studies have reported the analysis of human blood, serum, or urine, where better analytical characteristics of modified CPEs are necessary for the quantification of low amounts of the drug. However, only Farag [49] provided validation and comparison with the HPLC method. A good agreement of results was attained between the electrochemical and HPLC methods, although larger confidence intervals were reported in the first case.

Chitosan extracted from shrimp-shell waste with co-precipitated Fe_3O_4 nanocomposite [54] and chitosan hydrogel with TiO_2 nanotubes [55] were applied as CPE modifiers to enhance the electrochemical oxidation of diclofenac. Although the authors did not specify the way of electrode modification (bulk or surface layer) in the first case, both sensors exhibited high electrocatalytic activity towards the diclofenac oxidation, which was further supported by the adsorptive accumulation of the analyte into the chitosan matrix [55]. The latter work also compared and discussed in detail the differences in the construction between these two sensors, the effect of the electrochemical detection technique, and their impact on the resulting analytical performance [55].

For the electroanalysis of ibuprofen, modification with carbon nanomaterials has been employed not only to enhance the electron transfer of the carbon paste itself, but also to increase the available surface area, facilitating preconcentration or serving as a support for subsequent crafting, e.g., with Ni particles [56]. ZrO_2 nanopowder was

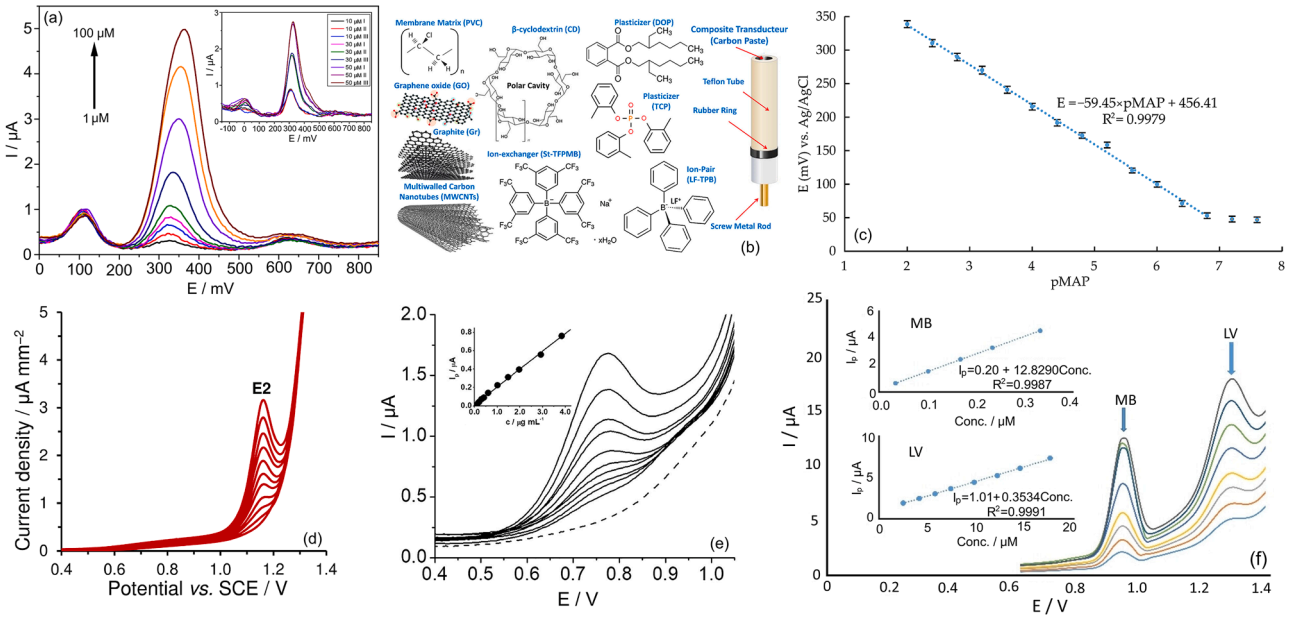


Fig. 2. Examples of the determination of various APIs at CPEs: (a) DPV of paracetamol in the presence of dopamine and tyrosine using the CPE modified with stevensite. The inset shows three spiked replicates of paracetamol in serum samples [47]; (b) Designing the potentiometric CP-sensor for levofloxacin [70]; (c) Calibration curve ($n = 11$ for each point) of the optimised CPE for potentiometric determination of maprotiline hydrochloride [76]; (d) SWV of 17β-estradiol at CPE with sodium dodecyl sulphate in 99 % acetonitrile containing 0.1 mol L^{-1} $LiClO_4$. Reprinted with permission from [77]; (e) SWV of azithromycin at AuNPs/CPE with the corresponding calibration (inset). Reprinted with permission from [42]; (f) Simultaneous DPV of mebendazole and levamisole hydrochloride at ZrO_2/CPE with the corresponding calibrations (inset). Reprinted with permission from [78].

also directly admixed with carbon paste to prepare simple ibuprofen sensors with an enhanced electrocatalytic effect [57]. The common excipients in tablets did not interfere with electrochemical detection, but insufficient selectivity was observed in the simultaneous determination of ibuprofen with caffeine, which is also present in some pharmaceutical formulations. The same modification approach was employed in the development of CPE for tramadol detection, utilising CuO nanoparticles prepared by co-precipitation in the presence of an extract from the leaves of *Origanum majorana* (acting as both a reductant and a stabiliser) in conjunction with MWCNTs [58].

Alteration of the carbon paste electrode surface is also possible via in situ modification with a suitable surfactant. Kamenická et al. [59] demonstrated on the detection of flufenamic acid that the addition of cetyltrimethylammonium bromide (CTAB) into the supporting electrolyte enlarges the surface of CPE by partial dissolution of the binder, exposing more graphite particles — via so-called erosion effect [60, 61]—, thus forming more possible sites for the accumulation or interaction with the analyte at the electrode/solution interface. The same positive effect of CTAB was observed in the analysis of structurally similar mefenamic acid at glucose-modified CPE. However, renewal of the electrode surface after each measurement was necessary to overcome electrode fouling [45]. Various modification approaches can also be purposely combined, as shown in the electrochemical detection of naproxen [62]. Firstly, the bulk modification of the carbon paste with graphene quantum dots was performed. Then, the electrode surface was coated with a film of poly(reduced-*o*-nitrobenzoic acid) by electropolymerisation after 8 cycles from 0.8 V to –1.5 V in 0.1 mol L^{-1} phosphate-buffered saline (PBS). These studies illustrate and confirm the flexibility and unique character of this heterogeneous electrode material, which is applicable with several advantages also in pharmaceutical analysis.

2.2. Antibiotics

Several recent reports have been published on the electrochemical detection and quantification of macrolide antibiotics. For example, Vajdle et al. [42] used a very simple modification of CPE by drop-coating the surface with a suspension of gold nanoparticles (AuNPs). After optimising the experimental conditions, especially the pH of the supporting electrolyte, the concentrations of azithromycin, clarithromycin, erythromycin ethylsuccinate, and roxithromycin were determined in sub-micromolar ranges. Comparison with HPLC-DAD was also provided for the analysis of azithromycin and roxithromycin in tablets, showing good agreement of the results. In other applications for azithromycin detection, CPEs were bulk-modified with graphene nanoribbons-$CoFe_2O_4$@NiO nanocomposite and 1-hexyl-3-methylimidazolium hexafluorophosphate as an ionic liquid binder [63] or coated with poly(threonine) by electropolymerisation of L-threonine in PBS (pH 9.0) using 30 potential cycles between −0.6 and 2.3 V [64]. Although both modifications resulted in an extended linear range of detection, no significant improvement was observed in other analytical parameters.

The compound containing Cu (Ca_2CuO_3), prepared using the coprecipitation method from eggshell waste, was incorporated into a CPE for simultaneous electrochemical detection of the fluoroquinolone antibiotics ciprofloxacin and ofloxacin [65]. The same inexpensive and widely available raw material, rich in calcium, served as a source for the synthesis of the chitosan-CaO nanocomposite. After mixing with graphite and paraffin wax, the resulting carbon paste-like electrode was used to monitor the antibiotic ornidazole by DPV in cathodic mode [66]. In this work, the limit of detection presented for ornidazole was relatively low compared to other electrochemical sensors based on modified glassy carbon or carbon paste electrodes; however, it was calculated using a signal-to-noise approach, which may yield artificially lower values. Similar mixed oxide materials containing Co were also used as bulk modifiers of other CPEs; either alone, as Co_3O_4 for the detection of levofloxacin [67], or in conjunction with graphene as $CoFe_2O_4$ for simultaneous determination of penicillin G and cephalexin [68].

CPEs can also be utilised as potentiometric sensors, where the ionophore is embedded directly in the electrode material or in the surface membrane, along with other components, such as a polymeric matrix, plasticisers, ionic sites, etc. A similar simple sensing principle is based on the formation of ion pairs between a usually cationic analyte

Table 1
Overview of electroanalytical methods using CPEs developed for the determination of APIs since 2020.

API	Sensor	Technique	Working medium	Linear range (mol L^{-1})	LOD (mol L^{-1})	Recovery (%)	Interfering compounds	Real sample	Ref.
Analgesics									
Acetaminophen	CPE-$MMTK_{10}$	DPV	0.1 mol L^{-1} PB (pH 6.5)	$1 \times 10^{-6} - 1.5 \times 10^{-5}$	5×10^{-7}	94	UA	—	[46]
Acetaminophen	CPE/MWCNTs	DPV	0.1 mol L^{-1} BRB (pH 7)	$5 \times 10^{-7} - 8 \times 10^{-5}$	1.5×10^{-5}	110	—	—	[41]
Acetaminophen	CCPE	DPV	0.1 mol L^{-1} PB (pH 6)	$1 \times 10^{-6} - 1 \times 10^{-3}$	8×10^{-8}	95	DA, AA, RS, SA, Ph	Human blood	[88]
Acetaminophen	CPE/SWCNTs	CV	0.2 mol L^{-1} PB (pH 7.2)	$1 \times 10^{-3} - 4 \times 10^{-3}$	5×10^{-6}	—	None	—	[48]
Acetaminophen	CPE/V_2O_5	AD	0.1 mol L^{-1} PB (pH 5)	$2 \times 10^{-6} - 8 \times 10^{-5}$	6×10^{-8}	—	None	Tablets, children sirup	[50]
Acetaminophen	CPE/RGC	DPV	0.5 mol L^{-1} H_2SO_4	$2.5 \times 10^{-6} - 1 \times 10^{-3}$	2×10^{-6}	—	None	Tablets	[51]
Acetaminophen	CPE/PYPX4R	DPV	0.2 mol L^{-1} PB (pH 7.4)	$1 \times 10^{-5} - 6 \times 10^{-5}$	8×10^{-8}	99–101	None	Tablets	[52]
Acetaminophen	CPE/Stevensite	DPV	0.1 mol L^{-1} PB (pH 6.7)	$6 \times 10^{-7} - 1 \times 10^{-4}$	2×10^{-8}	84–107		Tablets, human serum	[47]
Acetaminophen	CPE/rGO	SWV	0.1 mol L^{-1} AcB (pH 5)	$4 \times 10^{-6} - 2 \times 10^{-4}$	3×10^{-7}	99	None	Tablets, human urine	[49]
Aspirin	CPE/MWCNTs	DPV	0.1 mol L^{-1} BRB (pH 7)	$5 \times 10^{-7} - 8 \times 10^{-5}$	1×10^{-5}	110	—	—	[41]
Diclofenac	CPE/MWCNTs	DPV	0.1 mol L^{-1} BRB (pH 7)	$5 \times 10^{-7} - 8 \times 10^{-5}$	1×10^{-5}	110	—	—	[41]
Diclofenac	CPE-MCN	DPV	0.1 mol L^{-1} BRB (pH 3)	$2.5 \times 10^{-8} - 4 \times 10^{-6}$	7×10^{-9}	97–102	None	Tablets, human serum	[54]
Diclofenac	CPE/TiO_2/CS	DPV	0.1 mol L^{-1} PB (pH 5)	$2 \times 10^{-7} - 1 \times 10^{-5}$	1×10^{-8}	95–97	None	Tablets, synthetic urine	[55]
Flufenamic acid	CPE/CTAB	SWV	0.1 mol L^{-1} PB (pH 7)	$1 \times 10^{-6} - 1 \times 10^{-5}$	5.5×10^{-9}	—	—	—	[59]
Ibuprofen	CPE/MWCNs-Ni	SWV	0.1 mol L^{-1} AcB (pH 4.5)	$2.5 \times 10^{-5} - 5 \times 10^{-4}$	4×10^{-7}	94–98	None	Tablets, syrup	[56]
Ibuprofen	CCPE	DPV	0.1 mol L^{-1} PB (pH 6)	$1 \times 10^{-6} - 1 \times 10^{-3}$	4×10^{-8}	98	DA, AA, RS, SA, Ph	Human blood	[88]
Ibuprofen	CPE/ZrO_2	DPV	0.1 mol L^{-1} AcB (pH 4.5)	$2 \times 10^{-5} - 2.5 \times 10^{-4}$	4×10^{-6}	98–102	—	Tablets	[57]
Naproxen	CPE/MWCNTs	DPV	0.1 mol L^{-1} BRB (pH 7)	$5 \times 10^{-7} - 8 \times 10^{-5}$	9×10^{-6}	110	—	—	[41]
Naproxen	CPE/GQDs/PRNBA	DPV	0.1 mol L^{-1} PB (pH 7)	$1 \times 10^{-6} - 1 \times 10^{-4}$	7×10^{-8}	97–102	None	—	[62]
Mefenamic acid	CPE-GLU-CTAB	SWV	0.1 mol L^{-1} PB (pH 5)	$5 \times 10^{-8} - 5 \times 10^{-6}$	1×10^{-9}	96–99	None	Tablets, human urine	[45]
Methadone	MCPE	SWV	0.04 mol L^{-1} BRB (pH 8)	$1 \times 10^{-8} - 8 \times 10^{-6}$	3×10^{-9}	—	None	Human urine	[89]
Morphine	MCPE	SWV	0.04 mol L^{-1} BRB (pH 8)	$5 \times 10^{-9} - 2 \times 10^{-6}$	2×10^{-9}	—	None	Human urine	[89]
Tramadol	CPE/MWCNT-CuONPs	SWV	0.1 mol L^{-1} PB (pH 7)	$5 \times 10^{-8} - 2 \times 10^{-4}$	2.5×10^{-8}	96–104	—	—	[58]
Tramadol	CPE/SFOHNRs	DPV	0.1 mol L^{-1} AmB (pH 8)	$5 \times 10^{-7} - 7 \times 10^{-5}$	6.5×10^{-8}	98	None	Human serum and urine	[90]
Antibiotics									
Amoxicillin	CPE-CTAB	CV	0.1 mol L^{-1} PB (pH 6.5)	$1 \times 10^{-5} - 2 \times 10^{-4}$	6×10^{-6}	96–98	—	—	[91]
Azithromycin	CPE/HMIM PF_6/$CoFe_2O_4$@NiO-GN	SWV	0.1 mol L^{-1} PB (pH 7)	$1 \times 10^{-5} - 2 \times 10^{-3}$	7×10^{-7}	95–97	None	Capsules, urine	[63]
Azithromycin	CPE-PT	SWV	0.1 mol L^{-1} PB (pH 7.4)	$9 \times 10^{-6} - 1 \times 10^{-3}$	3×10^{-7}	99–103	None	Capsules	[64]
Azithromycin	CPE/AuNPs	SWV	0.1 mol L^{-1} BRB (pH 12)	$2 \times 10^{-7} - 3 \times 10^{-6}$	2×10^{-7}	—	OMA	Capsules	[42]
Cephalexin	CPE/$CoFe_2O_4$/Gr	DPV	0.1 mol L^{-1} PB (pH 8)	$1 \times 10^{-7} - 6 \times 10^{-5}$	2×10^{-8}	98–101	None	Honey, milk, tablets, human serum	[68]
Ciprofloxacin	CPE/Ca_2CuO_3	DPV	0.1 mol L^{-1} PB (pH 4)	$5 \times 10^{-8} - 1 \times 10^{-6}$	1×10^{-8}	99–100	None	Tablets, human serum	[65]
Clarithromycin	CPE/AuNPs	SWV	0.1 mol L^{-1} BRB (pH 12)	$6 \times 10^{-6} - 1 \times 10^{-5}$	2×10^{-6}	—	OMA	—	[42]
Erythromycin	CPE/AuNPs	SWV	0.1 mol L^{-1} BRB (pH 8)	$7 \times 10^{-7} - 6.5 \times 10^{-6}$	2×10^{-7}	—	OMA	—	[42]
Levofloxacin	CPE-PVC/LF-TPB	PM	0.1 mol L^{-1} AcB (pH 4)	$1 \times 10^{-5} - 1 \times 10^{-2}$	1×10^{-5}	93–101	Al^{3+}	Tablets, serum, urine	[70]
Levofloxacin	CPE-PG	DPV	0.2 mol L^{-1} PB (pH 4)	$3 \times 10^{-4} - 9 \times 10^{-4}$	8×10^{-7}	100	None	Tablets	[92]

(*continued on next page*)

Table 1 (*continued*)

API	Sensor	Technique	Working medium	Linear range (mol L^{-1})	LOD (mol L^{-1})	Recovery (%)	Interfering compounds	Real sample	Ref.
Levofloxacin	CPE/Co_3O_4	SWV	0.1 mol L^{-1} BRB (pH 5)	1×10^{-6} – 9×10^{-5}	4×10^{-7}	94–106	OA, UA	Tap water	[67]
Penicillin G	CPE/$CoFe_2O_4$/Gr	DPV	0.1 mol L^{-1} PB (pH 8)	5×10^{-7} – 5×10^{-5}	3×10^{-8}	—	None	Honey, milk, tablets, human serum	[68]
Ofloxacin	CPE/Ca_2CuO_3	DPV	0.1 mol L^{-1} PB (pH 4)	1×10^{-7} – 7×10^{-6}	3×10^{-8}	97–99	None	Tablets, human serum	[65]
Ornidazole	CPE/NC-CaO-CS	DPV	0.1 mol L^{-1} PB (pH 7)	2×10^{-8} – 5×10^{-6}	4×10^{-9}	96–102	ND	Milk, tablets	[66]
Roxithromycin	CPE/AuNPs	SWV	0.1 mol L^{-1} BRB (pH 12)	1×10^{-6} – 8×10^{-6}	3×10^{-7}	—	OMA	Tablets	[42]
Tetracycline	CCPE	SWV	0.1 mol L^{-1} AcB (pH 5)	5×10^{-7} – 8×10^{-7}	5×10^{-9}	—	—	—	[93]
Anticancer drugs									
Buserelin	CPE	DPV	0.5 mol L^{-1} AcB (pH 5.25)	6×10^{-6} – 1×10^{-4}	7×10^{-7}	97–103	—	Human serum	[75]
5-Fluorouracil	CPE/CeO_2.CuO	DPV	0.2 mol L^{-1} PB (pH 7)	1×10^{-7} – 4×10^{-5}	1×10^{-8}	94–99	—	Human urine	[71]
Gemcitabine	CPE/MWCNTs/AuNPs/ ssDNA	DPV	0.1 mol L^{-1} AcB (pH 4.5)	1×10^{-6} – 1×10^{-5}	5×10^{-7}	98–101	AA, FA	Human serum	[74]
Imatinib	CPE/ Fe_3O_4@MWCNTs@PANNFs	DPV	0.1 mol L^{-1} BRB (pH 6)	2×10^{-9} – 9×10^{-7}	4×10^{-10}	94–104	M, G, D, S	Human urine	[43]
Methotrexate	CPE/Ni-Co-LDH/IL	DPV	0.1 mol L^{-1} PB (pH 7)	2×10^{-8} – 1×10^{-4}	6×10^{-9}	99–104	—	Tablets, urine	[73]
Ribociclib	CPE/BPAC/$CoFe_2O_4$/ CuCoFe-MOF	DPV	0.1 mol L^{-1} PB (pH 1)	2×10^{-7} – 1×10^{-5}	2.5×10^{-8}	100–101	none	Human urine, tablets	[72]
Hormonal drugs									
17β-estradiol	CPE/BMI PF_6-Fe_3O_4NPs	SWV	0.1 mol L^{-1} BRB (pH 12)	1×10^{-7} – 1×10^{-6}	5×10^{-8}	—	None	Tablets, gel	[79]
17β-estradiol	CPE/SDS	SWV	0.1 mol L^{-1} $LiClO_4$ in MeCN	5×10^{-7} – 1×10^{-4}	2×10^{-7}	102	None	Tablets	[77]
Estriol	CPE/BMI PF_6-Fe_3O_4NPs	SWV	0.1 mol L^{-1} BRB (pH 11)	1×10^{-6} – 1×10^{-5}	3×10^{-7}	—	None	Tablets, cream	[79]
Estriol	CPE/OXL-9	CV	0.2 mol L^{-1} PB (pH 6)	4×10^{-5} – 1×10^{-4}	5×10^{-6}	—	None	—	[80]
Antidiabetic drugs									
Metformin	CPE/BNC/CuO/Cu_2O/ AgNPs	CV	0.04 mol L^{-1} BRB (pH 10)	1×10^{-7} – 1×10^{-3}	4×10^{-8}	98–101	None	Tablets, human urine	[85]
Antidepressants									
Amitriptyline	CPE-OPDC	DPV	0.1 mol L^{-1} PBS (pH 7)	1×10^{-8} – 9×10^{-4}	2×10^{-9}	98	None	Tablets, human serum and urine	[44]
Atomoxetine	CPE/FeONPs	DPV	0.1 mol L^{-1} BRB (pH 3)	2×10^{-4} – 3×10^{-2}	5×10^{-5}	99–100	None	Capsules, human plasma and urine	[81]
Maprotiline	CPE-PVC	PM	0.01 mol L^{-1} AcB (pH 4.5)	2×10^{-7} – 1×10^{-2}	1×10^{-7}	—	None	Tablets, human urine	[76]
Venlafaxine	CPE-OPDC	DPV	0.1 mol L^{-1} PBS (pH 7)	1×10^{-8} to 9×10^{-4}	3.5×10^{-9}	99	None	Tablets, human serum and urine	[44]
Anthelminthics									
Levamisole	CPE/ZrO_2NPs	SWV	0.04 mol L^{-1} BRB (pH 6)	8×10^{-7} – 2×10^{-5}	4×10^{-7}	100	None	Tablets, human plasma	[78]
Mebendazole	CPE/ZrO_2NPs	SWV	0.04 mol L^{-1} BRB (pH 6)	3×10^{-7} – 5×10^{-7}	1×10^{-8}	100	None	Tablets, human plasma	[78]
Anti-inflammatory drugs									
Mesalazine	CPE-PCR	DPV	0.2 mol L^{-1} PB (pH 7.4)	8×10^{-5} – 2×10^{-4}	1×10^{-7}	—	None	Tablets	[94]
Antiglaucoma drugs									
Dorzolamide	CPE/DRZ-PM/silica or β-CD	PM	Water solutions (pH 4.5–8.5)	1×10^{-6} – 1×10^{-2}	—	—	—	Eye drops	[82]

(*continued on next page*)

Table 1 (*continued*)

API	Sensor	Technique	Working medium	Linear range (mol L^{-1})	LOD (mol L^{-1})	Recovery (%)	Interfering compounds	Real sample	Ref.
Antivirals									
Acyclovir	CPE/CdO/Fe_3O_4	DPV	0.1 mol L^{-1} PB (pH 4)	$1 \times 10^{-6} - 1 \times 10^{-4}$	3×10^{-7}	94–104	None	Tablets, human plasma and urine	[87]
Anti-uric drugs									
Probenecid	CPE-PEDT	PM	0.1 mol L^{-1} BRB (pH 5)	$1 \times 10^{-6} - 1 \times 10^{-2}$	8×10^{-7}	99	None	Tablets, human plasma	[83]
Gastrokinetic drugs									
Mosapride citrate	CPE/ZrO_2	DPV	0.2 mol L^{-1} PB (pH 5)	$8 \times 10^{-8} - 1.5 \times 10^{-6}$	2×10^{-8}	100	None	Tablets, human plasma	[86]
Parkinson's drugs									
THP	CPE/THP-TPB/rGO	PM	0.1 mol L^{-1} AcB (pH 4)	$4 \times 10^{-7} - 1 \times 10^{-2}$	2.5×10^{-7}	97–102	None	Tablets, human urine	[84]

Abbreviations: AA; ascorbic acid, AD; amperometric detection, BMI.PF_6; 1-butyl-3-methylimidazolium hexafluorophosphate, BNC; bacterial nanocellulose composite, BPAC; banana-peel activated carbon, CCPE; clay modified carbon paste electrode, β-CD; β-cyclodextrin, CS; chitosan, CTAB; cetyltrimethyl ammonium bromide, D; dexamethasone, DA; dopamine, DRZ-PM; dorzolamide-phosphomolybdate complex, FA; folic acid, FeONPs; iron oxide nanoparticles, G; gefitinib, GLU; glucose, Gr; graphene, GQDs; graphene quantum dots, HMIM PF_6; 1-hexyl-3 methylimidazolium hexafluorophosphate, IL; ionic liquid, LDH; layered double hydroxide nanosheets, LF-TPB; levofloxacinium–tetraphenylborate ion pair, M; metoclopramide, MCN; magnetic chitosan nanocomposite, MCPE; magnetic carbon paste electrode, $MMTK_{10}$; montmorillonite, MOF; metal-organic framework, MWCNTs; multi-walled carbon nanotubes, ND; nitroimidazole derivatives, NPs; nanoparticles, NS; nanosilica, OA; oxalic acid, OMA; other macrolide antibiotics, OPDC; octahedral palladium-doped cobaltite composite, OXL-9; octoxynol-9, PANNFs; polyacrylonitrile nanofibers, PbA; Prussian blue analogue, PCR; polymerized-Congo red, PEDT; poly(3,4-ethylenedioxythiophene, PG; poly-glycine, pH; phenol, PM; potentiometry, PMR; polymurexide, PRNBA; poly(reduced-*o*-nitrobenzoic acid), PT; poly-threonine, PYPX4R; poly(yellow PX4R), RGC; regranulated/thermally treated cork, rGO; reduced graphene oxide, RS; resorcinol, S; Sunitinib, SA; salicylic acid, SDS; sodium dodecyl sulphate, SFOHNRs; SnO_2/α-Fe_2O_3 hierarchical nanorods, SWCNTs; single-walled carbon nanotubes, THP; trihexyphenidyl hydrochloride, THP-TPB; trihexyphenidyl-tetraphenylborate ion pair, and UA; uric acid.

and a large organic anion, followed by the subsequent extraction of the lipophilic ion pair into the bulk of carbon paste [69]. An ion pair can also be added to the sensing membrane, which covers the base CPE, or to the bulk of the electrode. An illustrative example is the detection of levofloxacin at several potentiometric CPE sensors made of various components. Ion pair of levofloxacin and tetraphenylborate and different amounts of modifiers, such as several carbonaceous materials (graphene, graphene oxide, reduced graphene oxide, and MWCNTs), polyvinyl chloride (PVC), tricresyl phosphate, and dioctyl phthalate as plasticisers, sodium tetrakis(trifluoromethyl)phenyl borate as an ion exchanger, and β-cyclodextrin, were mixed to prepare potentiometric CPEs or used as a membrane coating on the electrode surface. A major drawback of these potentiometric sensors is the limited stability of the membranes and a lifetime of only five days in the case of CPE plasticised with PVC [70].

2.3. Anticancer drugs

Bulk modification of CPEs with mixed oxide materials could also be used in the voltammetric determination of various anticancer drugs, utilising mainly their anodic oxidation in buffered media. The CeO_2–CuO nanocomposite was added during the preparation of the bare carbon paste to develop a sensor applicable to 5-fluorouracil [71]. Ghapanvari et al. synthesised Fe_3O_4 nanoparticles supported on the surface of MWCNTs, which were subsequently anchored onto polyacrylonitrile nanofibers in order to enhance the electrocatalytic behaviour of the modified CPE in the determination of imatinib [43]. This resulted in a very wide linear range with a remarkable detection limit of 0.4 nmol L^{-1}, which is the lowest value reported for any API reviewed in this article. In another application, banana-peel activated carbon served as a highly porous support for the anchoring of cobalt ferrite ($CoFe_2O_4$) and CuCoFe-metal-organic framework (MOF) to significantly enhance the electrochemical oxidation of ribociclib at modified CPE. Additionally, the authors evaluated the greenness and blueness of the proposed method using AGREE, BAGI, and GAPI indicators [72].

Similarly to mixed oxides, layered double metal hydroxides were also utilised for the bulk modification of CPE, e.g., Ni-Co layered double hydroxide nanosheets in conjunction with 1-butyl-3-methylimidazolium hexafluorophosphate as ionic liquid binder (together with liquid paraffin) for the determination of methotrexate [73]. An electrochemical DNA-biosensor for the selective determination of gemcitabine based on modified CPE was described in [74]. MWCNTs were mixed with basic components of carbon paste (graphite powder and paraffin oil), and AuNPs were formed on the surface of CPE with the aid of electrodeposition. Finally, a droplet of single-stranded DNA solution was cast on the surface of the electrode. An improved analytical performance was achieved compared to that of other DNA-based electrochemical sensors for gemcitabine; however, the stability and lifetime of the proposed sensor were not mentioned.

The last example of CPE sensors for anticancer drugs clearly demonstrates that there is not always an urgent need for the incorporation of structurally complex (nano)composites with high electrocatalytic activity whose synthesis may be complicated, costly, and time-consuming. Especially when various pharmaceutical formulations with a high content of APIs and with almost no electrochemical interference from other compounds are to be analysed. A simple bare CPE was used for the determination of buserelin at micromolar concentrations and was easily applied to the human serum assay [75].

2.4. Hormonal drugs

Klikarová et al. [77] tested, among other carbon-based electrodes, a non-aqueous CPE modified with sodium dodecyl sulphate for the simultaneous determination of estrogens, namely 17β-estradiol or

synthetic 17α-ethinylestradiol, with dienogest in pure acetonitrile. Analysed hormones are lipophilic; therefore, the use of acetonitrile leads to a significant simplification of the sample preparation, and interference from water-soluble compounds is minimised. Comparison with the reference HPLC method was also provided, showing good agreement of results for model and real samples, but higher confidence intervals in the case of nonaqueous CPE. Moreira et al. [79] used Fe_3O_4 nanoparticles and 1-butyl-3-methylimidazolium hexafluorophosphate ionic liquid for step-by-step bulk modification of a CPE for the detection of 17β-estradiol and estriol. No significant differences were found in the precision and accuracy of the analysis between the modified CPE and UV/VIS spectrometry. For estriol, a surface modification of CPE with drop-cast octoxynol-9 surfactant has also been reported [80].

2.5. Antidepressants

Simultaneous determination of amitriptyline and venlafaxine was achieved at CPE with octahedral Pd^{2+}-doped Co_3O_4 composite, prepared by the hydrothermal method. Although there was a partial overlap of DPV signals, very good recovery and repeatability were achieved for both pharmaceuticals in human serum and urine [44]. CPEs modified with several metal oxides, namely CuO, FeO, and ZnO, were tested for the enhanced electrocatalytic detection of atomoxetine, when FeO had exhibited the highest electroactivity among the selected compounds [81]. A potentiometric CPE sensor for the selective detection of maprotiline hydrochloride, based on the formation of ion pairs, was also reported [76]. Several ion-pairing agents, along with various binders and ionic salts, were tested, and CPE containing an ion pair with tetraphenylborate provided optimal response characteristics. Simpler preparation was noted over PVC membrane-based potentiometric sensors while maintaining high selectivity.

2.6. Other active pharmaceutical ingredients

Several potentiometric sensors that rely on chemically modified CPEs have recently been developed to monitor drugs from other classes of APIs. Silica and β-cyclodextrin, together with dorzolamide-phosphomolybdate ion exchanger, were incorporated into the CPE matrix for the determination of dorzolamide, an ophthalmic glaucoma agent, by direct potentiometry or potentiometric titration [82]. For the detection of probenecid, an uricosuric agent, poly(3,4-ethylenedioxythiophene) (PEDOT) dispersion and a PVC-based sensing membrane cocktail were drop-cast stepwise on the surface of CPE with the addition of MWCNTs [83]. The ion-selective principle, based on the formation of ion pairs, was also utilised in the construction of a potentiometric carbon paste sensor for the determination of trihexyphenidyl hydrochloride, an anticholinergic antiparkinsonian agent. Trihexyphenidyl-tetraphenylborate ion pair, together with reduced graphene oxide and other compounds, was admixed into the carbon paste matrix, and ratios of all components were tested for direct potentiometry of trihexyphenidyl hydrochloride. Although a very fast response and a sensor lifetime up to 20 days were achieved, limited stability of the sensing membrane was observed in prolonged assays [84].

Metal oxides were also used as modifiers of CPEs for the electrochemical detection of other drugs. Metformin, an antidiabetic agent, was determined at the CPE with copper oxides and silver nanoparticles (AgNPs). Firstly, bacterial nanocellulose was mixed with graphite powder and Nujol oil. Then, copper nanoparticles were electrodeposited under a constant potential, followed by electrochemical oxidation via cyclic voltammetry in 0.1 mol L^{-1} NaOH to form copper oxides. In the final step, AgNPs were electrodeposited over the modified surface of the CPE using the double potential step technique. The authors reported a low detection limit and an extended linear range compared to other electrochemical sensors for metformin, but the way to calculate the LOD was not specified [85]. CPEs with MWCNTs, ZrO_2, and SiO_2 nanoparticles were evaluated for the voltammetric detection of the anthelmintics mebendazole and levamisole hydrochloride. Modification with ZrO_2 was found to be optimal for the simultaneous quantification of both drugs. Full validation of the electrochemical method was provided, and no significant differences in results were found compared to the reference RPHPLC method [78]. The same authors also used the abovementioned CPE/ZrO_2 for the voltammetric determination of mosapride citrate, a gastroprokinetic agent, in the presence of cationic surfactant cetrimide (alkyltrimethylammonium bromide) [86]. In both works, the LODs and LOQs of the developed methods were determined by a visual inspection method, which is rather unusual. Finally, magnetic Fe_3O_4 nanoparticles were covered with CdO by precipitation and then embedded in the CPE for the determination of the antiviral drug acyclovir. Although good long-term stability of the current response was achieved, there was no significant improvement in analytical parameters over other electrochemical sensors [87].

2.7. Excipients

In accordance with standardised specifications of the European Pharmacopoeia, excipients also represent target substances in pharmaceutical analysis. Excipients in pharmaceuticals typically do not have a therapeutic effect; instead, they assist in the manufacturing, preparation, storage, and administration of drugs. As shown in Table 2, several excipients, such as adjuvants, colourants, flavouring agents, and vitamins, have been electrochemically determined at various types of modified CPEs. The electroanalytical methods developed over the past five years are critically evaluated below.

2.7.1. Colourants

First, it is essential to note that the presence of colourants in medications typically lacks therapeutic justification. One might think that colour has a justifiable role in helping patients distinguish between drugs when receiving multiple medications. However, this claim is now considered a historical relic. Today, drug manufacturers declare that when they run out of stock, they will replace controversial synthetic dyes with completely natural ones. Because most of the synthetic dyes used, especially sulfonated azo dyes [95], are electrochemically active substances, the azo group can be electrochemically reduced or oxidised. If a phenolic group is also present in the azo dye molecule, the anodic oxidation of the resulting phenolate in an alkaline environment may also be considered [96].

Herein, it should be noted that some pharmaceutical formulations may contain a mixture of several colourants to achieve the desired shade. Typical examples include throat lozenges such as Tantum Verde Orange and Honey 3 mg, which contain Quinoline Yellow (E104) and Sunset Yellow FCF (E110), or the variant Tantum Verde Lemon 3 mg, in which Sunset Yellow FCF is replaced by Indigo Carmine. Due to their distinct electrochemical properties, which reflect differences in molecular structure, the employment of CPEs for their simultaneous determination can be considered.

2.7.2. Flavouring agents

Among the excipients used in pharmaceutical preparations, flavouring agents play a key role not only by imparting flavour but also by improving palatability. These specific substances are commonly included in oral syrups, oral suspensions, elixirs, emulsions, lozenges, chewable tablets, effervescent tablets, dispersible tablets, and orally disintegrating tablets. Despite the currently preferred GC/MS analysis, voltammetric methods based on surfactant-modified carbon paste electrodes [97] may offer a more cost-effective alternative. However, it must be emphasised that electrochemical approaches cannot distinguish between structurally very similar compounds, such as vanillin and ethylvanillin. Therefore, the results always represent the sum of these substances, expressed as the equivalent of the most abundant form.

Table 2
Overview of electroanalytical methods using CPEs developed for determination of excipients in pharmaceutical preparations since 2020.

Excipient	Sensor	Technique	Working medium	Linear range (mol L^{-1})	LOD (mol L^{-1})	Recovery (%)	Interfering compounds	Ref.
Adjuvans								
Al^{3+}	CPE-Cu-MOFs	PM	HNO_3 with NaOH (pH 2 – 6.5)	$1 \times 10^{-7} - 1 \times 10^{-1}$	3×10^{-8}	99	None	[108]
Colourants								
Sunset Yellow FCF	CPE/MoO_2	SWV	0.1 mol L^{-1} PB (pH 2.1)	$1.6 \times 10^{-7} - 2 \times 10^{-6}$	6×10^{-8}	—	Allura red	[95]
Tartrazine	CPE/MoO_2	SWV	0.1 mol L^{-1} PB (pH 2.1)	$1.6 \times 10^{-7} - 2 \times 10^{-6}$	4×10^{-8}	—	None	[95]
Flavouring agents								
Vanillin	CPE/CTAB	DPV	0.1 mol L^{-1} PB (pH 7)	$8 \times 10^{-8} - 3 \times 10^{-3}$	2×10^{-10}	98–103	EVA	[97]
Vitamins								
VB2	CPE/FSN-Zonyl-Nafion	DPV	0.1 mol L^{-1} BRB (pH 8.2)	$1 \times 10^{-7} - 1 \times 10^{-4}$	3×10^{-11}	100–104	None	[103]
VB2	ShPE/MnO_2	DPV	0.1 mol L^{-1} BRB (pH 2)	$1 \times 10^{-7} - 1 \times 10^{-5}$	9×10^{-8}	—	—	[98]
VB6	CPE/$CuNP_S$	DPV	0.1 mol L^{-1} PB (pH 5)	$9 \times 10^{-6} - 1 \times 10^{-3}$	9×10^{-8}	98–103	None	[100]
VB6	CPE/$Fe_2O_3NP_S$	DPV	0.1 mol L^{-1} PB (pH 5)	$9 \times 10^{-6} - 1 \times 10^{-3}$	9×10^{-6}	97–103	None	[101]
VB9	CPE/$Ti(OH)_4$	DPV	0.1 mol L^{-1} PB (pH 6)	$4 \times 10^{-7} - 3 \times 10^{-5}$	1×10^{-9}	98–101	—	[102]
VB12	CPE/MnPF	SWV	0.1 mol L^{-1} PB (pH 7.6) with 0.03 mol L^{-1} KCl	$1 \times 10^{-11} - 1 \times 10^{-9}$	3×10^{-9}	95–105	None	[99]

Abbreviations: CTAB; cetyltrimethylammonium bromide, EVA; ethylvanillin, MeCN; acetonitrile, MnPF; manganese(II) polymeric film, MOFs; metal–organic framework, ShPE; shungite paste electrode, VB2; vitamin B2 (riboflavin), VB6; vitamin B6 (pyridoxine), VB9; vitamin B9 (folic acid), VB12; vitamin B12 (cobalamin).

2.7.3. Vitamins

The number of electrochemical methods proposed indicates that vitamins remain a topic of ongoing interest. Due to the mechanical instability of CPEs made of conventional graphites in organic solvents, voltammetric methods utilising CPEs modified with metal nanoparticles or powders of metal oxides have prevailed for the determination of water-soluble vitamins [98–103]. Regarding the voltammetric determination of lipophilic vitamins, two different approaches can be adopted:

- Extractive stripping voltammetry (ExSV) using a glassy carbon paste electrode (GCPE), which exploits its mechanical stability in aqueous-organic mixtures, where accumulation occurs by transfer of the analyte into a nonpolar binder [104].
- Direct voltammetry at non-aqueous carbon paste electrodes, modified with a surfactant, performed entirely in an organic medium [105].

2.7.4. Other excipients

Over the last five years, none of the already developed electroanalytical methods work has been devoted to the monitoring of amino acids contents in medications although some of them are part of various drug formulations as inevitable excipients, where they help improve the stability (Gly, His, and Pro), solubility (Gly) and efficacy of drugs (Arg and Gln) [106]. Also worth of mentioning are polysorbates and sodium laureth sulphate, an additive known as E 487, acting as stabilisers and emulsifiers in cream formulations. These additives can be determined by ion-pair formation-based potentiometric titration at modified carbon paste electrodes [107]. A very recent report highlighted that the content of the pasting liquid used in preparing these potentiometric sensors, specifically 2-nitrophenyl octyl ether (NPOE), depends entirely on the type of carbonaceous material [98].

2.8. Impurities

Quite surprisingly, until 2020, no scientific papers had been published on the use of carbon paste-based electrodes for the electrochemical determination of drug impurities, despite their importance and frequent presence in pharmaceutical preparations, where they can significantly affect the stability and efficacy of the drug or even pose a harmful effect on human health. Process-related impurities arising during the manufacturing, storage, or natural degradation of active pharmaceutical ingredients may include toxic heavy metals, carcinogenic residues of organic dyes and *N*-nitrosamines, and endocrine disruptors such as bisphenols, phthalates, parabens, and benzophenones, among others.

2.8.1. Heavy metals

Heavy metals may enter pharmaceutical products from various sources during the production process. These include raw materials, solvents, catalysts, and manufacturing equipment. Contamination occurring through packaging is also significant and must not be overlooked. Although heavy metals pose toxicological risks and must be strictly controlled, no electroanalytical method employing anodic stripping voltammetry at metal-film CPEs has been developed in the past five years. The explanation may lie in the strict purity requirements for input raw materials, as well as in the implementation of multi-element analysis using inductively coupled plasma mass spectrometry (ICP-MS).

2.8.2. Organic dyes

As mentioned in the previous chapter, synthetic dyes represent additives that are widely used to impart colour to pharmaceutical products. Nevertheless, many of these substances have already been prohibited by legislation, and some constitute contaminants that compromise health. Sudan dyes, such as Sudan I, may form as undesirable by-products in pharmaceuticals during the synthesis or degradation of permitted azo dyes, which contain the characteristic azo group. Although the extraction of non-polar Sudan I into the paste liquid [11,109] could have served to selectively distinguish it from other polar azo dyes with high sensitivity, this approach was, surprisingly, not applied. Indeed, it is precisely the high sensitivity of this approach that could facilitate the detection of even trace amounts of the carcinogenic compound Sudan I.

2.8.3. N-nitrosamines

Contamination with nitrosamines can occur during various stages of production, such as during the synthesis of pharmaceutical substances, where reactions between amines and nitrosating agents may take place. In this context, it can be noticed that certain medications, such as valsartan, ranitidine, and metformin, among others, have already been withdrawn from the market due to the presence of carcinogenic nitrosamines [110]. Moreover, these significant drug impurities, including *N*-nitrosodimethylamine (NDMA) [111], are electrochemically active and could also be determined at CPEs, similarly to the boron-doped diamond electrode (BDDE) [112].

2.8.4. Other significant impurities

Furthermore, *N*-nitrosamines that are detected in the class of sartan drugs, especially losartan, irbesartan (Avapro), and valsartan, have been identified as having organic azido contaminants, which are known to be highly reactive to DNA, leading to increased carcinogenic risk [113]. These azido impurities are formed during the preparation of the tetrazole moiety via the reaction of a nitrile intermediate with sodium azide. Since organic azides undergo electrochemical reduction at highly negative potentials ranging from −1.0 to −1.5 V vs. Ag/AgCl, depending on the substituents present and the selected pH [114], direct cathodic voltammetry at CPEs cannot be recommended due to the inability to fully remove oxygen from the electrode material.

A similar problem can be expected during the voltammetric determination of benzyl halides, which could also be electrochemically reduced in the same potential range [115]. Developing electroanalytical methods to monitor the levels of benzyl halides could be very important because these compounds are frequently used in the pharmaceutical synthesis of APIs, and their residual presence in medications is strictly regulated due to their potential carcinogenicity.

Furthermore, the active pharmaceutical ingredient can be, in itself, an unwanted contaminant in some multi-step syntheses or during the degradation of the final products. A typical example is the commonly used analgesic acetaminophen (APAP), which can contain hepatotoxic 4-aminophenol (4-AP) as a starting substance [116] along with diacetylated paracetamol formed as a byproduct during synthetic production. Unfortunately, some electroanalytical chemists have overlooked this fact in the development of heterogeneous sensors for the determination of acetaminophen.

3. Electrochemical mechanistic studies of pharmaceuticals at carbon paste electrodes

Nowadays, most scientific journals require a description of the electrode reaction mechanism employed for analytical purposes. However, few researchers optimise conditions to exploit it fully and avoid unnecessary modification [117]. From a pharmaceutical perspective, knowledge of the redox properties of all components in a drug formulation is much more important than merely quantifying the present API, because it can provide valuable insights into stability [118] and the potential toxicity of metabolites [96]. For this reason, molecular electrochemistry has its indispensable place in pharmaceutical analysis, and its possibilities with CPEs are critically evaluated below.

3.1. Influence of slow electrode reaction kinetics

In comparison with traditional metal electrodes, carbon electrodes have one undeniable advantage: their simultaneous application in both positive and negative potential ranges [119]. Unfortunately, it is also necessary to remember that for carbon pastes as heterogeneous electrode materials, the electrode reaction kinetics are typically up to two orders of magnitude slower than at solid carbon electrodes. As evidence, the values of heterogeneous rate constants (k^0) for ferricyanide determined by the Nicholson method [120] can be compared, namely 4.4×10^{-4}, 6.2×10^{-3}, and 4.3×10^{-2} cm s^{-1} for CPE, GCPE, and solid GCE, respectively [13].

Due to a slower charge transfer, a whole series of phenomena may arise that distort the electrochemical behaviour of the investigated substance. Typically, two single-electron reactions can often appear as one two-electron process. From an electroanalytical perspective, it is more advantageous to evaluate a single peak rather than two overlapping peaks. However, this approach is entirely inappropriate for describing the mechanism of the electrode reaction. Furthermore, the peaks tend to be much broader than on solid electrodes, which limits their applicability in the simultaneous determination of very similar compounds, especially trace amounts of toxic 4-aminophenol alongside acetaminophen.

Additionally, the extractive accumulation of less polar pharmaceutical compounds into the pasting liquid should not be overlooked. This phenomenon can also significantly complicate the interpretation of the resulting cyclic voltammograms. Very often, two electrochemical signals can be observed, although they correspond to a single electrode reaction. Specifically, they originate from the substance present in the diffusion layer and the same substance accumulated within the electrode body. Finally, the use of CPEs for studying the electrochemical behaviour of pharmaceuticals in strictly anhydrous media is only possible to a limited extent. This is because surfactants present in non-aqueous CPEs always contain an anionic or cationic head group in their structure, which can electrostatically influence the electrochemical behaviour [121].

On the other hand, the slow charge transfer can significantly assist in monitoring subsequent chemical and electrochemical reactions for substances governed by the ECE mechanism or its variants. For solid electrodes, slow scan rates are preferred to study subsequent reactions, allowing a sufficient number of products to form in the diffusion layer after the subsequent chemical reaction. The possible admixing of the investigated drug directly into the carbon paste can also lead to voltammetric measurements in the adsorbed state. This approach is carried out for solid electrodes by drop-casting, which can also be applied to CPEs.

3.2. Carbon paste electrode in spectroelectrochemical measurements

Although the use of various electrochemical techniques provides a detailed description of the electrochemical behaviour of the substances studied, it is not able to provide information about the structure of the resulting intermediates and products. Therefore, simultaneous employment of electrochemical and spectral techniques is required. Combining reaction-oriented electrochemistry with spectroscopy to identify chemical species in spectroelectrochemistry enables a more comprehensive analysis of single and multiple electron-transfer processes and redox reactions in general [122,123].

Unfortunately, there is no evidence that spectroelectrochemistry with a carbon paste electrode was used for studying pharmaceutical compounds. The preparation of a carbon paste in a thin layer that would be sufficiently conductive and electrically contactable represents a major limitation to its integration into spectroelectrochemical cells, which are often used in the molecular electrochemistry of pharmaceuticals. This arrangement enables the recording of UV–VIS-NIR, Raman, or IR spectra during voltammetric measurements, thereby allowing the identification of changes in functional groups that participate in the electrode reaction.

3.3. Investigation of electrochemical reaction pathways of pharmaceutical compounds

Despite the numerous disadvantages mentioned above, several scientific reports can be found on the use of CPEs in the molecular electrochemistry of drugs, as demonstrated in Table 3. Most studies on the electrochemical behaviour of drugs begin with theoretical molecular modelling, which involves performing quantum-chemical calculations

Table 3
Overview of mechanistic studies of selected pharmaceutical compounds.

API	Technique[1] + working electrode[2]	Studied properties / effect	Working medium[3]	Ref.
Acetaminophen	CV+PRGR/ CPE	Oxidation/ modelling to explain the redox reactive sites of PRGR	0.2 mol L^{-1} PB	[124]
Albendazole	CV + CPE	Oxidation/effect of scan rate, pH	BRB (pH 4.0)	[127]
Ceftriaxone	CV+ CPE/Au-BiNPs	Oxidation/ adsorption-diffusion mechanism	BRB (pH 4.0)	[128]
Edoxaban	CV, CA + CPE	Oxidation reaction mechanism, preparative electrolysis	BRB (pH 10.0)	[117]
Flutamide	DPV+ CPE/ DNA	Reduction/ interaction with DNA	0.3 mol L^{-1} PB (pH 7.0)	[129]
Furosemide	CV + CPE	Oxidation/effect of scan rate, pH	BRB	[133]
Itraconazole	CV + CPE	Oxidation/effect of scan rate, pH, adsorption	BRB	[134]
Lidocaine	CV + CPE	Oxidation/effect of scan rate	0.1 mol L^{-1} KNO_3	[135]
Mefenamic acid	CV + CPE/GO	Oxidation/effect of scan rate, temperature dependence	PB (pH 6.0)	[130]
Meloxicam	CV + CPE/GO	Oxidation/effect of scan rate, pH	—	[136]
Ruxolitinib	CV + CPE/poly (taurine)	Oxidation/effect of scan rate, pH	BRB (pH 4.7)	[131]
Tamoxifen	CV + CPE/ NiO	Oxidation/ electrode response kinetics	NaOH (pH 13)	[137]
Trimetazidine	CV + CPE	Oxidation/ electrostatic interaction/ PA	0.2 mol L^{-1} NaOH	[132]
Valocyclovir	CV + CPE	Oxidation/effect of scan rate, pH	PB (pH 7)	[138]
Warfarin	CV + CPE/ Ni_3CNPs	Oxidation/ interaction with K1	PB (pH 6)	[139]

Legend: 1) CV; cyclic voltammetry, DPV; differential pulse voltammetry, and CA; chronoamperometry; 2) Au-BiNPs; gold and bismuth nanoparticles, GO; graphene oxide, Ni_3CNPs; nickel carbide nanoparticles, NiO; nickel oxide, and PRGR; pyrogallol red 3) BRB; Britton-Robinson buffer, and PA; pre-anodisation.

such as density functional theory (DFT). It helps to understand the geometry of the molecule in the ground and excited states, and reveals the electron density within the molecule, as demonstrated, e.g., in reference [124].

Last but not least, the determination of products should be an integral part of the proper design of the electrode reaction mechanism; only in this way can one reliably say how the analysed substance is oxidised or reduced, as has been demonstrated, e.g., in [125,126]. In electroanalysis, where it is necessary to connect the observed signal with specific products of the electrochemical transformation of the analysed substance, the answer to this question is essential. Although proposals for these mechanisms appear in numerous publications, the path to them is quite diverse. Most often, a proposal for a mechanism can be found based on previously published works in which structurally similar substances have been analysed. This "spreads" across other publications where the electrode reaction mechanisms have only a theoretical character and have not yet been experimentally verified. In contrast, some publications propose mechanisms based on the experimental determination of the number of electrons combined with previously published reaction schemes, which can be considered a so-called "halfway" approach that does not necessarily yield correct results. Unfortunately, one can find questionable reports on the mechanism of electrochemical oxidation or reduction of pharmaceutical compounds at CPEs. For example, references [54,55] report on the mechanism of electrochemical oxidation of diclofenac in buffered aqueous media, but they propose different reaction pathways. Hence, the incorrect mechanism can appear in further publications.

Therefore, the first step in designing the mechanism should lead to determining which form of the substance in the solution undergoes oxidation (or reduction) under the given conditions, assuming that the investigated substance is stable in that solution. Quite often, the electrochemical properties of the substance investigated depend upon the pH, and thus its protonated or, vice versa, deprotonated form undergoes the redox transformation. This information can be obtained from the basic description of its electrochemical behaviour or from the optimisation of the parameters for an electroanalytical method. Furthermore, it is often necessary to consider when designing a new mechanism, which more-or-less stable intermediates (e.g., radicals) are formed, as they cannot be revealed by preparative electrolysis. However, the other tools of molecular (organic) electrochemistry mentioned above can be used for this purpose.

In the next section, selected publications (Table 3) on mechanistic studies of pharmaceuticals (drugs) are critically commented on. In most cases, such studies accompany the electroanalysis of pharmaceuticals, as is also the case in a study by Ganesh et al. [124] devoted to acetaminophen (paracetamol). In addition to the fundamental electrochemical study using cyclic voltammetry at a Pyrogallol Red-modified carbon paste electrode, density functional theory (DFT) was employed to elucidate the behaviour of the redox-active sites of the electrode modifier. Furthermore, the work by Mersal and Elrouby concerning albendazole [127] and ceftriaxone [128], respectively, is a typical example of a published mechanism based solely on the literature and basic experiments carried out just with cyclic voltammetry. In the latter case, the role of the modifier is not explained in the mechanistic study. In contrast, Rocha et al. [117] presented the mechanism of edoxaban oxidation at the bare CPE based not only on cyclic voltammetric measurements but also on bulk electrolysis followed by UHPLC/MS to identify the electrode reaction products, which is an example of a correct experimental approach. Another, rather exceptional work presented by Brahman et al. is a study on the interaction of flutamide with DNA bases [129]. The author has found that this drug preferentially interacts with the adenine and guanine of the DNA immobilised on the surface of CPE.

Moreover, in the work of Talikoti et al. [130] regarding the electrochemical behaviour of mefenamic acid at the CPE modified with graphene oxide, in addition to the theoretical treatment of voltammetric data, experiments were conducted to study the temperature dependence. Concerning the mechanism, it is claimed that a radical product is formed, but it is evident that the CV experiment shows a low stability of the oxidation product; hence, the technique for proving the existence of a radical should be used in this case. Ruxolitinib, the drug for myelofibrosis treatment, was studied by Subak et al. [131], who have shown characterisation of the surface of the polytaurine modifier by various methods (such as CV, EIS, SEM and FTIR). However, the effect of drug interaction with the surface of the modifier had been omitted, and the proposed mechanism was based solely on literature data. In a study by Fan et al. [132], the specially pre-anodised surface of the carbon paste electrode played a significant role in the electrostatic attraction between the protonated trimetazidine and the negatively charged electrode surface.

The comments mentioned above document a variety of methods applicable for studying the electrode reaction mechanisms of pharmaceutical compounds at carbon paste-based electrodes. Their specific features may bring both advantages and difficulties in some mechanistic

studies. The authors must be aware that not all published mechanisms are correct.

4. Electroanalysis of pharmaceuticals at carbon paste-based electrodes: A survey of the latest trends and some prospects

Despite a somewhat reduced selection of recent publications, it is believed that this compilation reflects the latest activities and achievements in the field, demonstrating that pharmaceutical analysis remains one of the leading areas of practical applications of CPEs and CMCPEs. Secondly, this review also documents a great diversity of newly published methods and procedures in their entirety, when covering all the main categories of pharmaceuticals. Third, the vitality of the field is underscored by the finding that recently published contributions encompass a diverse palette of new substances that had not been analysed in previous periods (it represents good several tenths even in a selection excluding CP-biosensors.)

Within the 2020s reviewed, evident trends and features can be observed, and, in comparison with other related areas, they also outline some prospects for the near future. Among such advances and implications, the following points can be highlighted:

- A great majority of proposed configurations are CMCPEs and CP-biosensors, where the respective modifiers have repeatedly been represented by various nanomaterials and inorganic-organic hybrids, often as new or previously unused products.
- On the other hand, traditional modifiers are still of interest. Among them, one can quote surfactants, ion-exchangers, cyclodextrin, or even some classical reagents, such as Congo red, Prussian blue, and Murexid, known from chelatometric titrations.
- As expected, new forms of carbon mainly nanotubes and graphene– are prevailing in the proposed configurations, which is a trend that can be observed among carbon paste-based electrodes already for two or three decades. In other words, there is a continuing decline in the use of traditional CPEs, as their employment is becoming rather rare.
- Among the latest trends that should be stated at first, it is undoubtedly the rapidly growing emphasis on environmentally friendly or, now more often, green aspects. In electroanalysis with carbon paste-based electrodes, one can reveal three main directions: (i) still more frequent use of natural materials from ecological/green renewable sources used for the preparation of new types of CPEs, (ii) choice of green procedures for synthesising new modifiers for CMCPEs, and (iii) evaluation of methods employing both CPEs and CMCPEs by means of specially proposed eco-scoring. Regarding the first category, biochar [140] or shungite (mineralised carbon) [98] have recently been tested as environmentally friendly alternatives to graphite moieties. Additionally, some vegetable oils [141], or the latest hit among green pasting liquids – natural deep eutectic solvents (NADES) [142] – have been of interest as new liquid binders. Among green modifiers, the above-mentioned biochar can also be used [143,144]. Interesting natural materials used as modifying agents are bamboo charcoal [145] or banana-peel activated carbon [72]. Other reports also include some metal-oxide nanoparticles among the green modifiers [54,146]. A far largest group of green-assessed methods employing various CMCPEs and carbon paste-like composites is represented by reports where the authors have been using various green-profile evaluations, such as *Eco-Scale* score, *Green Analytical Procedure Index* (GAPI), *Complex-GAPI, Analytical Greenness* (AGREE), *Blue Applicability Grade Index* (BAGI) or *National Environmental Methods Index* (see e.g. [72,86,147–151]. In all these assessments, CPEs and CMCPEs are primarily integral parts of the respective analytical procedures, and their contribution to the overall eco-score varies from case to case.
- Some association with green principles is being attributed to wearable electronics as another attractive topic in contemporary instrumental analysis. Among electrochemical sensors and biosensors of this kind, one can find the first examples of constructions that utilise special carbon pastes (e.g., [152]), when two configurations have been proposed for pharmaceutical and clinical analysis [153,154]. In fact, these rather scarce reports in the otherwise wide family of other wearable sensors correspond to the fact that carbon paste, as a soft and non-compact material, is not very suitable for such configurations.
- Similarly, one would also anticipate some new entries for in vivo monitoring, especially if mini- and micro-CPEs had already been successfully applied in this way in the early era of carbon pastes [23, 155]. However, in the latest period, the only example of such an in vivo sensor has been found employing a Bi-plated carbon nanotube paste (mini)electrode in the chronoamperometric mode for medical diagnostics of epinephrine [156].
- Finally, perhaps the hottest trend in the current period is the phenomenon of artificial intelligence (AI) and its utilisation in instrumental analysis. Regarding the combination of carbon paste electrodes, pharmaceutical analysis, and AI, there is no report to date that connects these three topics. However, there are already first reports of employing AI in environmental analysis, food quality control, or clinical diagnostics [157–159]. And because the corresponding electrochemical measurements in such attempts have been performed with carbonaceous electrodes, it can be believed that any AI-assisted experiment with a true carbon paste-based sensor will commence soon.

5. Conclusion

This article reviews the activities in pharmaceutical (electro)analysis with electrodes and sensors based on carbon paste, with a focus on the latest achievements in the 2020s. The respective publications in a representative selection primarily concern the development and proposals of electroanalytical methods/procedures for determining APIs in real samples, mostly in commercial formulations. In contrast, theoretically oriented studies on electrode reaction mechanisms from the same period (overviewed in Section 3) represent only a minor part, which may be explained by the fact that carbon paste-based electrodes are preferred for practical analysis, whereas they are less suitable for more theoretical investigations.

It can be concluded that pharmaceutical analysis with CPEs and CMCPEs, as well as with CP-biosensors (not covered herein), is still a vital area due to the extraordinary versatility of carbon pastes in terms of their numerous configurations and variable modifications, both hand in hand with a fine compatibility with the latest ecological demands on a green electrode material. New publications in the field appear regularly, as evidenced by the most recent entries from 2024 to 2025 [100,149, 160–167]. Thus, the authors of this text consider such continuing popularity as a sign of a stable position in today's electroanalysis, and they are similarly optimistic about further progress in the future.

CRediT authorship contribution statement

Radovan Metelka: Writing – original draft, Investigation, Conceptualization. **Tomáš Mikysek:** Writing – original draft, Investigation. **Ivan Švancara:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Milan Sýs:** Writing – original draft, Supervision, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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